Integrating PfSense with Wazuh SIEM

Introduction

A **firewall** is a critical security solution that inspects and controls incoming and outgoing traffic on a network or device. Think of it like a **security guard at the entrance of a building**: it decides who can enter and who must stay out, based on predefined rules.

In modern environments, firewalls help:

- Prevent unauthorized access
- Block malicious traffic
- Segment networks for better security

There are different types of firewalls: **packet-filtering, stateful inspection, proxy-based, and next-generation firewalls (NGFWs)**, each providing increasing levels of inspection and control.

In this guide, we build a **hands-on SOC lab** using:

- **pfSense** → an open-source firewall and router.
- **Wazuh** → an open-source SIEM for log analysis, alerting, and threat detection.

By integrating pfSense logs into Wazuh, we simulate **real-world SOC operations**, monitoring both network and endpoint events.

1 Step 1: Requirements

Before starting, ensure you have the following:

1. **Virtual Machine Software** → VMware Workstation Pro (used in this guide, but VirtualBox also works).
2. **pfSense ISO** → Download the latest version from the official pfSense website.
3. **Wazuh Manager** → Running and accessible in your lab environment.

2 Step 2: Creating a Virtual Machine for pfSense

2.1 Open VMware

Navigate to **File ¿ New Virtual Machine**.

2.2 Select the Installation Media

Choose **Installer disc image file (ISO)** and browse for the downloaded pfSense ISO.

2.3 Name Your VM

Use a descriptive name such as **pfSense** and pick a save location.

2.4 Allocate Disk Space

- **20 GB minimum** (store as a single file recommended).

2.5 Customize Hardware

- **RAM**: 1 GB minimum (2 GB recommended).
- **CPU**: 1 vCPU (2 cores recommended).
- **Network Adapters**:
  - **Adapter 1 (WAN)** → NAT (simulates internet access).
  - **Adapter 2 (LAN)** → Host-only (isolated internal lab network).

2.6 Assign Interfaces

- **WAN (em0)** → Configured via DHCP (IP in 192.x.x.x/24 range).
- **LAN (em1)** → Static IP 192.168.1.1/24.

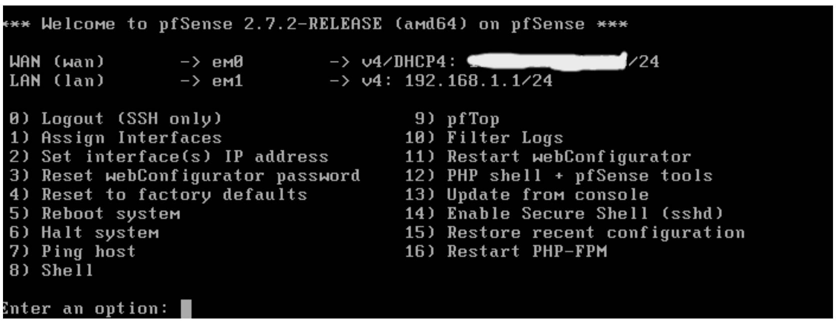

Figure 1: pfSense interface

3 Step 3: Accessing the pfSense Web Interface

1. Open a browser from a VM (e.g., Kali Linux) connected to the LAN network.
2. Go to: https://192.168.1.1/
3. Log in with default credentials:
   - Username: `admin`
   - Password: `pfsense`

Important: Change the default credentials immediately after login.

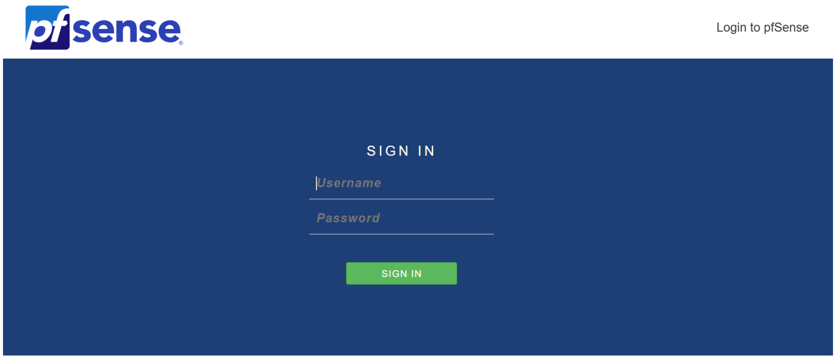

Figure 2: pfSense web interface login

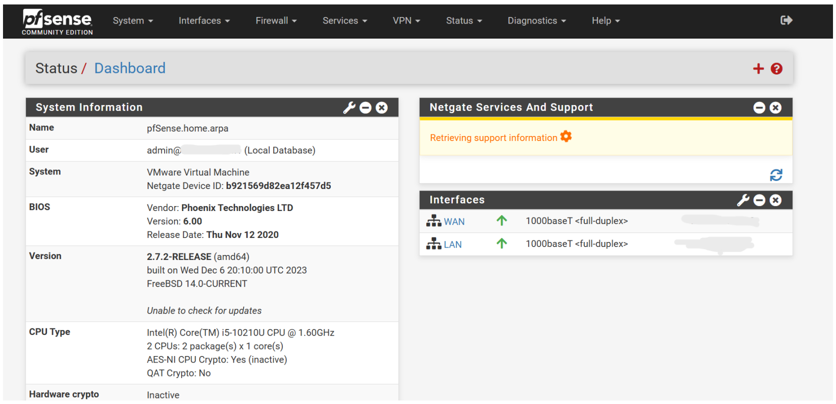

Figure 3: pfSense dashboard

4 Step 4: Why Combine pfSense with Wazuh?

- **pfSense** acts as the **network security gateway**, generating logs on traffic, connections, and blocked events.
- **Wazuh** functions as the **Security Information and Event Management (SIEM)** platform, collecting and analyzing logs to detect suspicious activity.

Together, they form a **mini SOC environment**, where firewall events are monitored and correlated with endpoint activity.

5 Step 5: Configuring pfSense Logs in Wazuh

5.1 Enable Remote Logging in pfSense

- Go to **Status** → **System Logs** → **Settings**.
- Check **Enable Remote Logging**.
- Add your **Wazuh server's IP** (e.g., 192.168.200.50) under **Remote Syslog Servers**.
- Set **Port** to 514 (default syslog).
- Select log categories to forward: System, Firewall, VPN, DHCP, DNS.

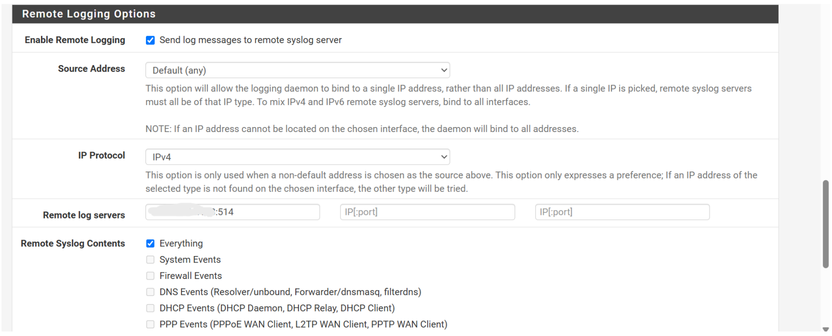

Figure 4: pfSense remote logging configuration

5.2 Configure Wazuh Syslog Input

On the Wazuh Manager, edit `/var/ossec/etc/ossec.cfg`:

```
<ossec_config>
  <!-- pfSense syslog input -->
  <remote>
    <connection>syslog</connection>
    <port>514</port>
    <protocol>udp</protocol>
    <allowed-ips>YOUR_PF_SENSE_IP</allowed-ips>
    <local_ip>YOUR_WAZUH_SERVER_IP</local_ip>
  </remote>
</ossec_config>
```

Restart Wazuh after changes.

6 Step 6: Creating a Custom Decoder and Rules for pfSense Logs

Since pfSense logs have a unique format, Wazuh may not parse them correctly out of the box.

6.1 Custom Decoder

A **decoder** extracts key fields (source IP, destination IP, protocol).

Example (`pfsense-custom-decoder.xml`):

```
<decoder name="pfsense-custom">
  <prematch>filterlog</prematch>
</decoder>

<decoder name="pfsense-fields">
  <parent>pfsense-custom</parent>
  <regex>(\w+)\d+:\S*,\S*,\S*,(\S+),\S*,\S*,(\S+),\S*,\S*,\S
     *,\S*,\S*,\S*,\S*,\S*,\S*,\S*,\S*,(\S+),\S*,(\S+),(\S+),(\d
     +);(\d+),\S*</regex>
  <order>logsource,id,section,protocol,srcip,dstip,srcport,
     dstport</order>
</decoder>
```

6.2 Custom Rules

A **rule** assigns severity levels and alerts based on matched events.

Example (`pfsense-custom-rules.xml`):

```xml
<group name="pfsense,firewall,">

  <!-- Generic pfSense log -->
  <rule id="100111" level="3">
    <decoded_as>pfsense-custom</decoded_as>
    <description>pfSense log received</description>
  </rule>

  <!-- Successful login -->
  <rule id="100112" level="3">
    <if_sid>100111</if_sid>
    <match>Successful login</match>
    <description>pfSense: Successful login from $(srcip)</description>
  </rule>

  <!-- Allowed traffic -->
  <rule id="100113" level="5">
    <if_sid>100111</if_sid>
    <match>pass</match>
    <description>pfSense: Allowed traffic from $(srcip) to $(dstip)</description>
  </rule>

  <!-- Blocked traffic -->
  <rule id="100114" level="7">
    <if_sid>100111</if_sid>
    <match>block</match>
    <description>pfSense: Blocked traffic from $(srcip) to $(dstip)</description>
  </rule>

  <!-- Authentication error -->
  <rule id="100115" level="10">
    <if_sid>100111</if_sid>
    <match>authentication error</match>
    <description>pfSense: Authentication error for user $(dstuser) from $(srcip)</description>
  </rule>

</group>
```

- **Level 3 (Informational):** Successful logins.
- **Level 5 (Normal alert):** Allowed traffic.
- **Level 7 (Warning):** Blocked traffic.

- **Level 10 (High):** Authentication errors.

Conclusion

By combining **pfSense** and **Wazuh**, you now have a **practical SOC lab** that:

- Monitors and analyzes firewall traffic (allowed vs. blocked).
- Centralizes logs for correlation and alerting.
- Detects authentication attempts and suspicious behavior.
- Provides a foundation for hands-on practice in SOC operations.

This setup is ideal for **cybersecurity students, SOC analysts in training, or homelab enthusiasts** who want to explore **network security monitoring** in a safe environment.